# ECE 2305 – Introduction to C Programming

## Programming Tutorial 2

Write a C++ Program to investigate the Basic Data Types in C++. Declare a variable of type **int**, **double**, **char** and **bool** and assign a value to each of the variables. Determine the address of each variable using the **&** operator and the memory requirement using the **sizeof** keyword. Use the following bit of code as an example for type **int** data.

## Main Code

```
//ECE 2305 Tutorial 02 - Basic Data Types in C++
#include <iostream>

using namespace std;

int main() {
    cout << "C++ Data Types Exploration\n\n";

    int intValue = 50;
    cout << "Value of intValue: " << intValue << endl;
    cout << "Size (bytes): " << sizeof(intValue) << endl;
    cout << "Memory Address: " << &intValue << "\n\n";

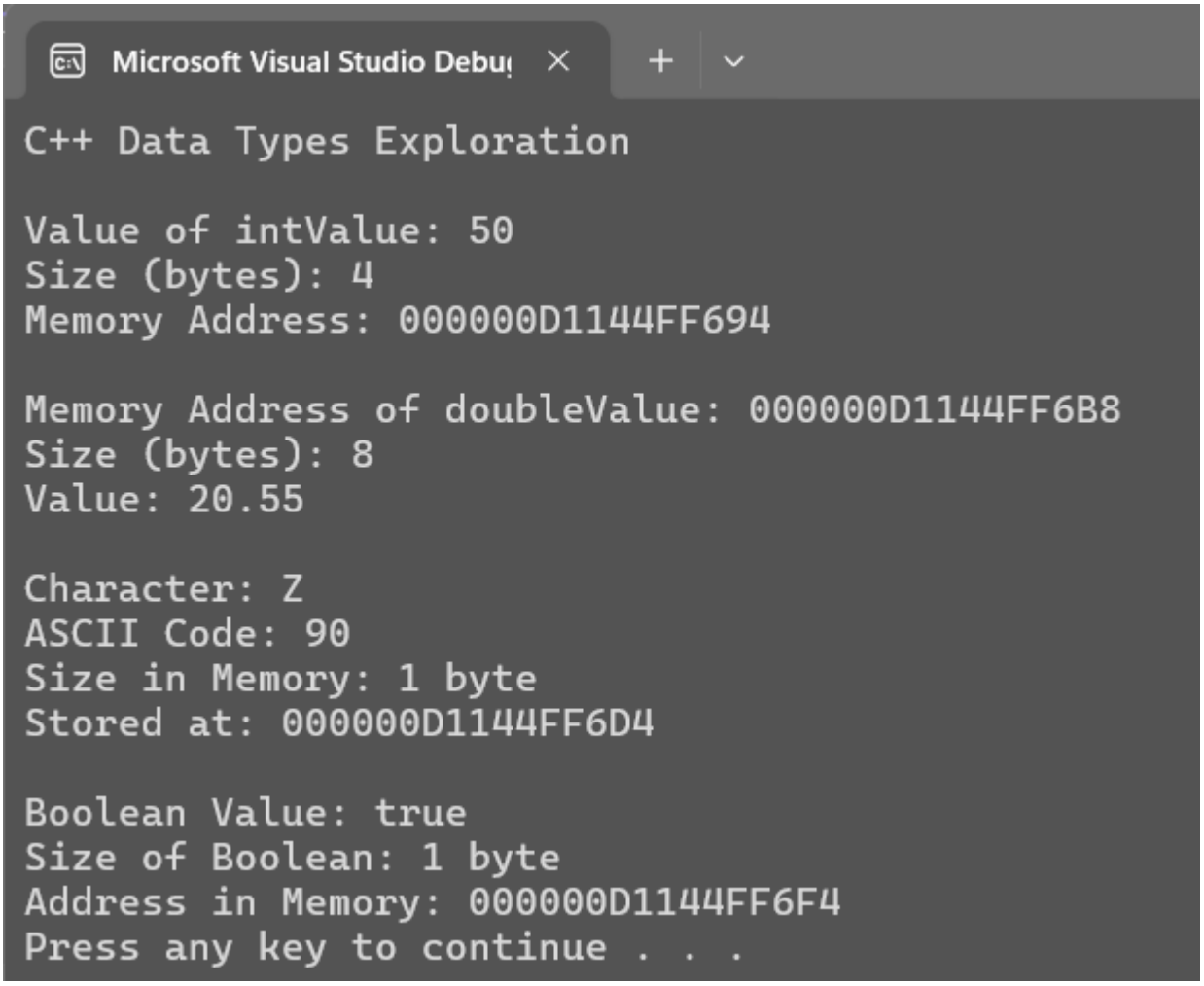
    double doubleValue = 20.55;
    cout << "Memory Address of doubleValue: " << &doubleValue << endl;
    cout << "Size (bytes): " << sizeof(doubleValue) << endl;
    cout << "Value: " << doubleValue << "\n\n";

    char charValue = 'Z';
    cout << "Character: " << charValue << endl;
    cout << "ASCII Code: " << int(charValue) << endl;
    cout << "Size in Memory: " << sizeof(charValue) << " byte" << endl;
    cout << "Stored at: " << (void*)&charValue << "\n\n";

    bool boolValue = true;
    cout << "Boolean Value: " << boolalpha << boolValue << endl;
    cout << "Size of Boolean: " << sizeof(boolValue) << " byte" << endl;
    cout << "Address in Memory: " << &boolValue << endl;

    system("pause");
    return 0;
}
```

# Output


Microsoft Visual Studio Debu
C++ Data Types Exploration
Value of intValue: 50
Size (bytes): 4
Memory Address: 000000D1144FF694
Memory Address of doubleValue: 000000D1144FF6B8
Size (bytes): 8
Value: 20.55
Character: Z
ASCII Code: 90
Size in Memory: 1 byte
Stored at: 000000D1144FF6D4
Boolean Value: true
Size of Boolean: 1 byte
Address in Memory: 000000D1144FF6F4
Press any key to continue . . .